Fiscal Year:	FY 2025	**Task Last Updated:**	FY 05/19/2026
PI Name:	Sargsyan, Ashot M.D.		
Project Title:	Imaging-Based Human Physiology Research and Innovation in Early Spaceflight: Inspiration4 Mission		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77058	**Congressional District:**	36
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	TRISH--EXPAND
Start Date:	05/01/2021	**End Date:**	09/30/2025
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	SpaceX		
Flight Assignment:	SpaceX Inspiration4 NOTE: End date changed to 09/30/2025 per TRISH (Ed., 8/7/24). NOTE: End date changed to 04/30/2024 per TRISH (Ed., 8/4/22). NOTE: End date changed to 04/30/2023 per TRISH (Ed., 8/4/22). Project formerly titled, "Miniaturized Medical Imaging for Space: Butterfly iQ+ Ultrasound".		
Key Personnel Changes/Previous PI:	Douglas Ebert, Ph.D. has been added as a Co-Investigator (Ed., 1/6/24).		
COI Name (Institution):	Ebert, Doug Ph.D. (KBR)		
Grant/Contract No.:	NNX16AO69A-EXP0002		
Performance Goal No.:			
Performance Goal Text:	NOTE: End date changed to 09/30/2025 per TRISH (Ed., 8/7/24).		

Task Description:	EXPAND Project This proposal aims to perform a series of low-resource, high-yield human research objectives using a hand-held ultrasound imaging system Butterfly iQ+ (Butterfly Network, USA) during the commercial mission Inspiration4: 1) urinary bladder function in microgravity; 2) major vein and fluid status evaluation; 3) ocular imaging self-surveillance protocol; and 4) gallbladder function in microgravity. These objectives address recognized concerns for space flight health, with parallel consideration of benefits to certain patient populations, including pediatric malignancies. They are transferable, scalable, and will establish precedents for long-term inquiries. Traditional imaging-based research relies heavily on crew training, infrastructure, and ground personnel; it foregoes science opportunities that require quick deployment, simple procedures, flexible schedules, and crew autonomy. This proposal leverages the inherent features of the Butterfly iQ+ ultra-compact ultrasound imager, the team's experience with hybrid crew skill management solutions, and the nature of the Inspiration4 mission.
Rationale for HRP Directed Research:	
	Results generated during these experiments will benefit the realms of 1) spaceflight physiology and 2) skill management. Physiological elements of the project will add to space medicine and physiology knowledge base by providing data about 1) urinary bladder function, 2) internal jugular vein flow patterns, and 3) ocular structure changes. The data collected during this project will be unique given that they are collected in the early phases of spaceflight exposure (first 3 days), as compared with the majority of available data which have been collected much later in flight (30+ days for ISS missions).
Research Impact/Earth Benefits:	In the skill management realm, this project will have application for remote medicine, spaceflight medicine, home care (telemedicine), and general medical training literature. Skill management solutions will be developed and testing that will directly benefit programs requiring that minimally-trained operators perform complex tasks and are particularly applicable to situations where communication is limited either in quality or quantity. Latency and bandwidth considerations are central to exploration spaceflight and a variety of Earth based scenarios where increased autonomy is necessary. This work contributes to the academic knowledge base in the training literature but will simultaneously be building operational experience and products for both NASA and commercial space operators.
Task Progress:	This summary consists of three sections: A. Original Objectives of Inspiration4 Mission; B. Ultrasound Innovation - Polaris Dawn; C. Butterfly iQ+ Antarctic Program. Projects A and B (Inspiration4, Sep. 2021, and Polaris Dawn 2024) used procedures to study physiological changes in early days of microgravity, and to evaluate success of autonomous imaging operation with procedures of various complexity. These investigations chart a path to assured autonomous imaging in future exploration missions. A. Original Objectives of EXP-0002 (Inspiration4 Mission) This project performed low-resource human research on known concerns for spaceflight health, using a Butterfly iQ+ handheld ultrasound system (Butterfly Network, USA) during the Inspiration4: 1) urinary bladder function in microgravity; 2) major vein evaluation with impedance breathing; and 3) ocular imaging self-surveillance protocol. With minimal training (by proxy) and full crew autonomy, just-in-time (JIT) instructions provided the conceptual and procedural information. To counteract bias, quality assessment was conducted upon receipt of imaging data to include scoring each image for anatomical accuracy and technical quality on a scale of 0 - 3 with data usability threshold set at 2.0. Bladder imaging scores were 2.33 ± 0.24[SD]; (2.0 - 2.5). Internal jugular vein (IJV) images had highest scores (2.36 ± 0.51[SD]; 1.8 - 2.9) and showed no anomalies (thrombi, stasis, flow reversal, spontaneous echo contrast (SEC)), in contrast to International Space Station (ISS) data. Impedance Threshold Device (ITD) reduced right IJV area by 36% preflight (supine) and 29% inflight. Inflight ocular data scored below usability limit in all subjects (0.76 ± 0.51 [SD]; 0.0 – 1.1), and were deemed inadequate. ITDs were used by all crewmembers for hemodynamic modification during IJV testing (first time). Key results of the experiment are published. Ed. Note: See Bibliography. B. Ultrasound Innovation (Polaris Dawn mission) This extension study was similar to the original with similar approaches and pursued 1) Urinary bladder function; 2) IJVs with use of ITD; 3) Bubble detection/Venous Gas Embolism (VGE) evaluation; 4) Effects of lower extremity flow restriction on central and peripheral circulation (added in May 2024 as synergistic collaboration). Objectives 3 and 4 were not completed inflight for reasons beyond our control; data from Objective 1 and 2 were contiguous with Inspiration4 to generate larger data sets from early microgravity (combined analysis data pending). IJV evaluation relied on a procedure guide produced by Level Ex to innovate in procedural performance in autonomy. IJV data revealed only normal flow patterns with no anomalies. Bladder imagery was correctly timed and provided valuable insight into this important aspect of microgravity physiology. Summary results from flight projects (A and B): 1) The handheld system provided data of sufficient quality for routine and specialized measures pursued by this investigation. 2) IJV imaging attained good quality to allow analysis of structural and functional data. No flow anomalies were detected bilaterally in 7 subjects (8 crewmembers) during early microgravity exposure. 3) Impedance Threshold Device (ITD 7) breathing showed both preflight (supine) and inflight reduction of cross-sectional areas of IJVs bilaterally when applied, suggesting a potential for counteracting anomalous flow redistribution in long-duration missions. 4) Ocular procedure was deemed unsuccessful due to strict demands to gaze control and probe position (project A). For assured success, skill management solutions beyond linear JIT tools, as well as 2-person operation, are recommended. 5) Urinary bladder function analysis did not reveal any generalizable trends. The automated volume tool performed well

	and could be a valuable part of future medical capability. C. Analog: Butterfly iQ+ Antarctic Program. Exploring Autonomous Procedures - data collection and high-level analysis are complete. While similar in approach and technology, the small number of operators of Projects A and B could not provide definitive data on success rates and limitations of autonomous imaging in operational conditions. This project with the Australian Antarctic Division (AAD) addressed feasibility and management of autonomous imaging in a spaceflight analog environment of four isolated Antarctic stations. A large-scale human-in-the-loop imaging trial using Butterfly iQ+ was conducted to evaluate feasibility of 7 autonomous imaging procedures relevant to spaceflight needs: 1) eyes and orbits, 2) IJV and IVC, 3) kidneys, 4) urinary bladder, 5) heart, 6) arm vein, 7) vascular access phantom. Images and questionnaire results were analyzed from 210 imaging sessions completed in autonomy. Data analysis focused on quality of imagery collected by expeditioner operators: those with no medical experience, lay surgical assistants, and physicians. Effectiveness of autonomous protocols was evaluated using a data analysis framework with rigid rating criteria, using a scale of 0-3 for anatomical accuracy and technical quality. Procedure success scores were derived. Operator questionnaire responses (208) were a critical part of data that identified barriers to success. In addition to main results, the team provides imaging data analysis methodology that can benefit informed adoption of future autonomy solutions without undue bias. A reference terrestrial database of autonomously acquired normative data has been created. Further, the team's conclusions include the highly relevant nature of the AAD research stations as spaceflight analogue locations. Future studies should examine just-in-time training and guidance systems based on advanced digital technologies such as AI agents, which are likely to further improve success rates in autonomous imaging scenarios in both spaceflight and resource-limited ground locations.
Bibliography Type:	Description: (Last Updated: 05/19/2026)
Abstracts for Journals and Proceedings	Cherry J, Sargsyan A, Ebert D, Everson A, Urquieta E, Wu J, Ayton J. "Evaluation of autonomy enabling solutions for handheld ultrasound use in an Antarctic spaceflight analog." 76th International Astronautical Congress (IAC 2025), Sydney, Australia, Sep 29-Oct 3, 2025. Abstracts. 76th International Astronautical Congress (IAC 2025), Sydney, Australia, Sep 29-Oct 3, 2025. , Sep-2025